



Innovation & entrepreneurship hub for education rural youth(SURE Trust-IERY)

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G4-25 VLSI

Assignment-2

Project Title: Up/Down Counter using OpenLane Physical Design Flow:

About Up/Down Counter

An Up/Down counter is a sequential digital circuit that can:

- Increment count (UP mode)
- Decrement count (DOWN mode)

Working:

- Controlled by a direction signal (UP/DOWN)
- Changes value on every clock edge
- Uses flip-flops for state storage

	Clock	Mode	Output
↑		Up	0001 → 0010 → 0011
↑		Down	0011 → 0010 → 0001

Perform complete RTL-to-GDSII flow using OpenLane and analyze:

- Maximum Frequency
- Setup Uncertainty
- Derate Value
- Worst Slack
- Power
- Area
- Route Length
- IR Drop
- Utilization

Start OpenLane Container:

```
Last login: Sat Jun 6 14:58:01 on tty000
mounika@mounikas-MacBook-Air ~ % cd ~/OpenLane
mount
cd /Users/mounika/OpenLane && \
docker run --rm -v /Users/mounika/OpenLane/openlane:/Users/mounika/OpenLane/designs:openlane/install -v /Users/mounika:/Users/mounika --v /Users/mounika/.ciel:/Users/mounika/.ciel -e PDK_ROOT=/Users/mounika/.ciel -e PDK=sky130A --user 581:20 -ti ghcr.io/the-openroad-project/openlane:ff5509f65b17bfa4068d5336495ab1718987ff69-arm64v8
OpenLane Container (ff5509f):/openlane$
```

Enter OpenLane Interactive Mode:

```
OpenLane Container (ff5509f):/openlane$ ./flow.tcl -interactive
OpenLane v1.0.2 (ff5509f65b17bfa4068d5336495ab1718987ff69)
Copyright (c) 2020-2025 Efabless Corporation and contributors.
Available under the Apache License, version 2.0. See the LICENSE file for more details.

% package require openlane 0.9
version conflict for package "openlane": have 1.0.2, need 0.9
% █
```

PREPARING FOR DESIGN:

```
OpenLane Container (ff5509f):/openlane$ ./flow.tcl -interactive
OpenLane v1.0.2 (ff5509f65b17bfa4068d5336495ab1718987ff69)
Copyright (c) 2020-2025 Efabless Corporation and contributors.
Available under the Apache License, version 2.0. See the LICENSE file for more details.

% package require openlane 0.9
version conflict for package "openlane": have 1.0.2, need 0.9
% █
```

```
input clk
```

☐ rst up down

output reg [3:0] count

always @(posedge clk or posedge rst)

```
if(rst)
```

```
count <= 4'b0000:
```

— *John F. Kennedy*

$$\text{count} = \text{count} + 1; \quad \text{if } \text{count} \geq 100 \text{ then } \text{count} = 0;$$

endmodule

endmodule

Create config.json:

```
{  
  "DESIGN_NAME": "up_down_counter_32",  
  "VERILOG_FILES": "dir::src/up_down_counter_32.v",  
  "CLOCK_PORT": "clk",  
  "CLOCK_PERIOD": 6,  
  "FP_CORE_UTIL": 70,  
  "FP_PDN_CORE_RING": true,  
  "SYNTH_CLOCK_UNCERTAINTY": 0.6,  
  "SYNTH_TIMING_DERATE": 0.03,  
  "PL_TARGET_SKEW": 0.3  
}
```

[illegible]

B: Worst Slack at Each Stage

Synthesis:

```
mounika@mounikas-macbook-air RUN_2026.06.01_09.24.36 % cat reports/syntnesis/z-syn_sta.summary.rpt
=====
report_tns
=====
tns 0.00

=====
report_wns
=====
wns 0.00

=====
report_worst_slack -max (Setup)
=====
worst slack 6.91

=====
report_worst_slack -min (Hold)
=====
worst slack 0.32
mounika@mounikas-MacBook-Air RUN_2026.06.01_09.24.36 %
```

Placement:

```
mounika@mounikas-macbook-air RUN_2026.06.01_09.24.36 % cat reports/placement/i1-dpi_sta.summary.rpt
=====
report_tns
=====
tns 0.00

=====
report_wns
=====
wns 0.00

=====
report_worst_slack -max (Setup)
=====
worst slack 6.86

=====
report_worst_slack -min (Hold)
=====
worst slack 0.21
mounika@mounikas-MacBook-Air RUN_2026.06.01_09.24.36 %
```

CTS:

```
mounika@mounikas-MacBook-Air RUN_2026.06.01_09.24.36 % cat reports/cts/13-cts_sta.summary.rpt
```

```
=====
report_tns
=====
ns 0.00

=====
report_wns
=====
ns 0.00

=====
report_worst_slack -max (Setup)
=====
orst slack 6.83

=====
report_worst_slack -min (Hold)
=====
orst slack 0.20
mounika@mounikas-MacBook-Air RUN_2026.06.01_09.24.36 %
```

Routing:

```
mounika@mounikas-MacBook-Air RUN_2026.06.01_09.24.36 % cat reports/routing/z1-grt_sta.summary.rpt
```

```
=====
report_tns
=====
tns 0.00

=====
report_wns
=====
wns 0.00

=====
report_worst_slack -max (Setup)
=====
worst slack 6.89

=====
report_worst_slack -min (Hold)
=====
worst slack 0.31
mounika@mounikas-MacBook-Air RUN_2026.06.01_09.24.36 %
```

C:Power Analysis

Synthesis Power:

mounika@mounikas-MacBook-Air RUN_2026.06.01_09.24.36 % cat reports/synthesis/2-syn_sta.power.rpt

report_power

===== Typical Corner =====

Group	Internal Power	Switching Power	Leakage Power	Total Power (Watts)	
Sequential	1.86e-05	1.32e-05	4.26e-11	3.18e-05	67.1%
Combinational	1.01e-05	5.49e-06	7.80e-11	1.56e-05	32.9%
Macro	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Pad	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Total	2.87e-05	1.87e-05	1.21e-10	4.74e-05	100.0%
	60.6%	39.4%	0.0%		

mounika@mounikas-MacBook-Air RUN_2026.06.01_09.24.36 %

CTS POWER:

mounika@mounikas-MacBook-Air RUN_2026.06.01_09.24.36 % cat reports/cts/13-cts_sta.power.rpt

report_power

===== Typical Corner =====

Group	Internal Power	Switching Power	Leakage Power	Total Power (Watts)	
Sequential	1.96e-05	2.74e-06	4.66e-11	2.24e-05	20.6%
Combinational	6.15e-05	2.49e-05	2.04e-10	8.64e-05	79.4%
Macro	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Pad	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Total	8.11e-05	2.76e-05	2.50e-10	1.09e-04	100.0%
	74.6%	25.4%	0.0%		

mounika@mounikas-MacBook-Air RUN_2026.06.01_09.24.36 %

ROUTING POWER:

mounika@mounikas-MacBook-Air RUN_2026.06.01_09.24.36 % cat reports/routing/21-grt_sta.power.rpt

report_power

===== Typical Corner =====

Group	Internal Power	Switching Power	Leakage Power	Total Power (Watts)	
Sequential	1.97e-05	1.62e-06	4.66e-11	2.14e-05	19.6%
Combinational	6.46e-05	2.33e-05	2.13e-10	8.78e-05	80.4%
Macro	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Pad	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Total	8.43e-05	2.49e-05	2.59e-10	1.09e-04	100.0%
	77.2%	22.8%	0.0%		

mounika@mounikas-MacBook-Air RUN_2026.06.01_09.24.36 %

SIGNOFF POWER:

```
mounika@mounikas-MacBook-Air RUN_2026.06.01_09.24.36 % cat reports/signoff/31-rcx_sta.power.rpt

=====
report_power
=====
===== Typical Corner =====
=====
Group                Internal Power    Switching Power    Leakage Power    Total Power (Watts)
-----
Sequential           1.96e-05         2.49e-06           4.66e-11         2.21e-05         19.4%
Combinational        6.48e-05         2.73e-05           3.33e-10         9.21e-05         80.6%
Macro                0.00e+00         0.00e+00           0.00e+00         0.00e+00         0.0%
Pad                  0.00e+00         0.00e+00           0.00e+00         0.00e+00         0.0%
-----
Total                8.44e-05         2.98e-05           3.79e-10         1.14e-04         100.0%
                        73.9%           26.1%             0.0%
mounika@mounikas-MacBook-Air RUN_2026.06.01_09.24.36 %
```

D: Area Analysis

```
mounika@mounikas-MacBook-Air RUN_2026.06.01_09.24.36 % cat reports/synthesis/1-synthesis.AREA_0.stat.rpt

Printing statistics.

=== up_down_counter ===

Number of wires:          20
Number of wire bits:      23
Number of public wires:   4
Number of public wire bits: 7
Number of memories:       0
Number of memory bits:    0
Number of processes:      0
Number of cells:          20
  sky130_fd_sc_hd__and2b_2  2
  sky130_fd_sc_hd__dfrtp_2  4
  sky130_fd_sc_hd__inv_2    5
  sky130_fd_sc_hd__o21a_2   1
  sky130_fd_sc_hd__o21ai_2  1
  sky130_fd_sc_hd__or2b_2   2
  sky130_fd_sc_hd__xnor2_2  3
  sky130_fd_sc_hd__xor2_2   2

Chip area for module '\up_down_counter': 257.747200
mounika@mounikas-MacBook-Air RUN_2026.06.01_09.24.36 %
```

E: Route Length

```
mounika@mounikas-MacBook-Air RUN_2026.06.01_09.24.36 % find results -name "*.def"
results/cts/up_down_counter.def
results/floorplan/up_down_counter.def
results/final/def/up_down_counter.def
results/routing/up_down_counter.def
results/placement/up_down_counter.def
mounika@mounikas-MacBook-Air RUN_2026.06.01_09.24.36 %
```

F: IR Drop Analysis:

```
mounika@mounikas-MacBook-Air RUN_2026.06.01_09.24.36 % head -20 reports/signoff/32-1rdrop-VPWR.rpt
Instance name, X location, Y location, Voltage
input1, 36.895, 46.24, 1.79998
FILLER_0_11_69, 36.895, 40.8, 1.79998
FILLER_0_8_67, 36.895, 35.36, 1.79998
FILLER_0_9_72, 36.895, 35.36, 1.79998
FILLER_0_10_71, 36.895, 35.36, 1.79998
FILLER_0_6_65, 36.895, 29.92, 1.79996
FILLER_0_7_64, 36.895, 29.92, 1.79996
FILLER_0_7_72, 36.895, 29.92, 1.79996
FILLER_0_5_65, 36.895, 24.48, 1.79998
FILLER_0_2_65, 36.895, 19.04, 1.79998
FILLER_0_3_69, 36.895, 19.04, 1.79998
input2, 36.895, 13.6, 1.79998
FILLER_0_1_69, 36.895, 13.6, 1.79998
FILLER_0_11_51, 27.93, 40.8, 1.79998
FILLER_0_12_53, 27.93, 40.8, 1.79998
_29_, 27.93, 35.36, 1.79997
_33_, 27.93, 35.36, 1.79997
FILLER_0_9_49, 27.93, 35.36, 1.79997
_18_, 27.93, 29.92, 1.79995
mounika@mounikas-MacBook-Air RUN_2026.06.01_09.24.36 %
```

G: Utilization:

```
mounika@mounikas-MacBook-Air RUN_2026.06.01_09.24.36 % cat reports/floorplan/3-initial_fp_core_area.rpt
mounika@mounikas-MacBook-Air RUN_2026.06.01_09.24.36 %
5.52 10.88 41.4 46.24 %
mounika@mounikas-MacBook-Air RUN_2026.06.01_09.24.36 %
```

-
- synthesis: Converts Verilog → Gate level netlist
 - floorplan:
 - Defines chip
 - Sets IO pins
 - Core placement
 - placement: Places standard cells
 - CTS: Builds clock network
 - routing: Connects all wires
 - signoff:
 - Timing analysis
 - Final check

Implementation of RTL to GDSII Flow using OpenLane for Up/Down Counter Design

To implement the complete ASIC physical design flow using OpenLane and generate the final GDSII layout of an **up/down counter design**, and analyze the layout using KLayout.

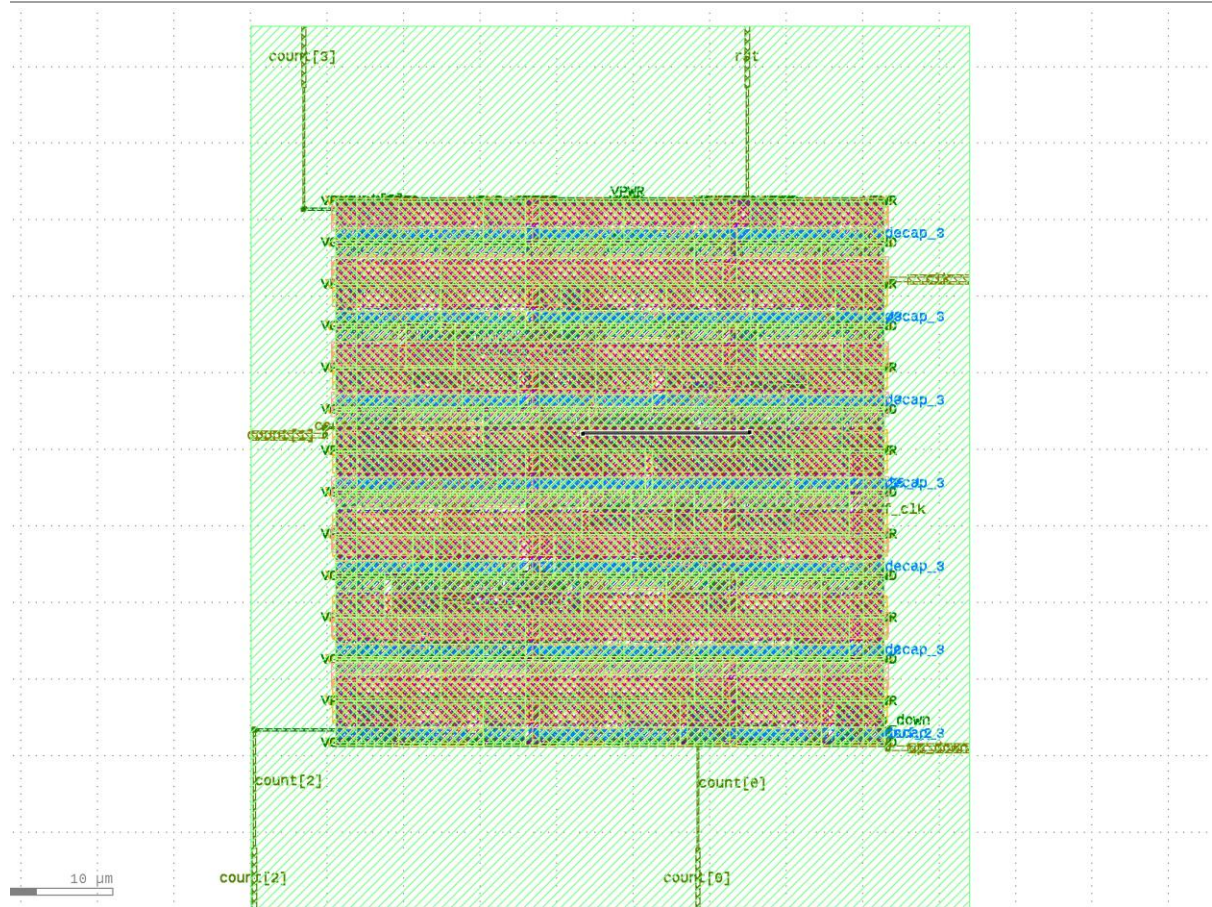
Tools Used

- OpenLane (v1.0.2)
- Docker

- Sky130 PDK
- KLayout (for layout visualization)
- macOS terminal

```
[STEP 30]
[INFO]: Running Multi-Corner Static Timing Analysis at the nom process corner (log: designs/up_down_counter/runs/RUN_2026.06.07.08.58.39/logs/signoff/30-rcx_msta.nom.log)...
[STEP 31]
[INFO]: Running Single-Corner Static Timing Analysis at the nom process corner (log: designs/up_down_counter/runs/RUN_2026.06.07.08.58.39/logs/signoff/31-rcx_sta.log)...
[WARNING]: Module sky130_fd_sc_hd__topvprvgsd.1 blackboxed during sta
[WARNING]: Module sky130_ef_sc_hd__decap.12 blackboxed during sta
[WARNING]: Module sky130_fd_sc_hd__f111.1 blackboxed during sta
[WARNING]: Module sky130_fd_sc_hd__f111.2 blackboxed during sta
[STEP 32]
[INFO]: Creating IR Drop Report (log: designs/up_down_counter/runs/RUN_2026.06.07.08.58.39/logs/signoff/32-ir_drop.log)...
[WARNING]: VDRM_LOC_FILES is not defined. The IR drop analysis will run, but the values may be inaccurate.
[STEP 33]
[INFO]: Running Magic to generate various views...
[INFO]: Streaming out GDSII with Magic (log: designs/up_down_counter/runs/RUN_2026.06.07.08.58.39/logs/signoff/33-gdsii.log)...
[INFO]: Generating MAGLEF views...
[INFO]: Generating lef with Magic (/openlane/designs/up_down_counter/runs/RUN_2026.06.07.08.58.39/logs/signoff/33-lef.log)...
[STEP 34]
[INFO]: Streaming out GDSII with KLayout (log: designs/up_down_counter/runs/RUN_2026.06.07.08.58.39/logs/signoff/34-gdsii-klayout.log)...
[STEP 35]
[INFO]: Running XDR on the layouts using KLayout (log: designs/up_down_counter/runs/RUN_2026.06.07.08.58.39/logs/signoff/35-xor.log)...
[INFO]: No XDR differences between KLayout and Magic gds.
[STEP 36]
[INFO]: Running Magic Spice Export from LEF (log: designs/up_down_counter/runs/RUN_2026.06.07.08.58.39/logs/signoff/36-spice.log)...
[STEP 37]
[INFO]: Writing Powered Verilog (logs: designs/up_down_counter/runs/RUN_2026.06.07.08.58.39/logs/signoff/37-write_powered_def.log, designs/up_down_counter/runs/RUN_2026.06.07.08.58.39/logs/signoff/37-write_powered_verilog.log)...
[STEP 38]
[INFO]: Writing Verilog (log: designs/up_down_counter/runs/RUN_2026.06.07.08.58.39/logs/signoff/37-write_powered_verilog.log)...
[STEP 39]
[INFO]: Running LVS (log: designs/up_down_counter/runs/RUN_2026.06.07.08.58.39/logs/signoff/39-lvs.lef.log)...
[STEP 40]
[INFO]: Running Magic DRC (log: designs/up_down_counter/runs/RUN_2026.06.07.08.58.39/logs/signoff/40-drc.log)...
[INFO]: Converting Magic DRC database to various tool-readable formats...
[INFO]: No DRC violations after GDS streaming out.
[STEP 41]
[INFO]: Running OpenROAD Antenna Rule Checker (log: designs/up_down_counter/runs/RUN_2026.06.07.08.58.39/logs/signoff/41-arc.log)...
[STEP 42]
[INFO]: Running Circuit Validity Checker ERC (log: designs/up_down_counter/runs/RUN_2026.06.07.08.58.39/logs/signoff/42-erc_screen.log)...
[INFO]: Saving current set of views in 'designs/up_down_counter/runs/RUN_2026.06.07.08.58.39/results/final'...
[INFO]: Saving runtime environment...
[INFO]: Generating final set of reports...
[INFO]: Created manufacturability report at 'designs/up_down_counter/runs/RUN_2026.06.07.08.58.39/reports/manufacturability.rpt'.
[INFO]: Created metrics report at 'designs/up_down_counter/runs/RUN_2026.06.07.08.58.39/reports/metrics.csv'.
[INFO]: There are no max slew, max fanout or max capacitance violations in the design at the typical corner.
[INFO]: There are no hold violations in the design at the typical corner.
[INFO]: There are no setup violations in the design at the typical corner.
[SUCCESS]: Flow complete.
[STEP 1]
[INFO]: Running Synthesis (log: designs/up_down_counter/runs/RUN_2026.06.07.08.58.39/logs/synthesis/1-synthesis.log)...
[STEP 2]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/up_down_counter/runs/RUN_2026.06.07.08.58.39/logs/synthesis/2-sta.log)...
[STEP 3]
[INFO]: Running Initial Floorplanning (log: designs/up_down_counter/runs/RUN_2026.06.07.08.58.39/logs/floorplan/3-initial_fp.log)...
[WARNING]: Current core area is too small for the power grid settings chosen. The power grid will be scaled down.
[INFO]: Floorplanned with width 35.88 and height 35.36.
[STEP 4]
[INFO]: Running IO Placement...
[STEP 5]
[INFO]: Running Tap/Decap Insertion (log: designs/up_down_counter/runs/RUN_2026.06.07.08.58.39/logs/floorplan/5-tap.log)...
[INFO]: Power planning with power (VPWR) and ground (VGND)...
[STEP 6]
[INFO]: Generating PDN (log: designs/up_down_counter/runs/RUN_2026.06.07.08.58.39/logs/floorplan/6-pdn.log)...
[STEP 7]
[INFO]: Running Global Placement (log: designs/up_down_counter/runs/RUN_2026.06.07.08.58.39/logs/placement/7-global.log)...
[STEP 8]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/up_down_counter/runs/RUN_2026.06.07.08.58.39/logs/placement/8-gpl_sta.log)...
[STEP 9]
[INFO]: Running Placement Resizer Design Optimizations (log: designs/up_down_counter/runs/RUN_2026.06.07.08.58.39/logs/placement/9-resizer.log)...
[STEP 10]
[INFO]: Running Detailed Placement (log: designs/up_down_counter/runs/RUN_2026.06.07.08.58.39/logs/placement/10-detailed.log)...
[STEP 11]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/up_down_counter/runs/RUN_2026.06.07.08.58.39/logs/placement/11-dpl_sta.log)...
[STEP 12]
[INFO]: Running Clock Tree Synthesis (log: designs/up_down_counter/runs/RUN_2026.06.07.08.58.39/logs/cts/12-cts.log)...
[STEP 13]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/up_down_counter/runs/RUN_2026.06.07.08.58.39/logs/cts/13-cts_sta.log)...
[STEP 14]
[INFO]: Running Placement Resizer Timing Optimizations (log: designs/up_down_counter/runs/RUN_2026.06.07.08.58.39/logs/cts/14-resizer.log)...
[STEP 15]
[INFO]: Running Global Routing Resizer Design Optimizations (log: designs/up_down_counter/runs/RUN_2026.06.07.08.58.39/logs/routing/15-resizer_design.log)...
[STEP 16]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/up_down_counter/runs/RUN_2026.06.07.08.58.39/logs/routing/16-rsz_design_sta.log)...
[STEP 17]
[INFO]: Running Global Routing Resizer Timing Optimizations (log: designs/up_down_counter/runs/RUN_2026.06.07.08.58.39/logs/routing/17-resizer_timing.log)...
[STEP 18]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/up_down_counter/runs/RUN_2026.06.07.08.58.39/logs/routing/18-rsz_timing_sta.log)...
[STEP 19]
[INFO]: Running Global Routing (log: designs/up_down_counter/runs/RUN_2026.06.07.08.58.39/logs/routing/19-global.log)...
[INFO]: Starting OpenROAD Antenna Repair Iterations...
[STEP 20]
[INFO]: Writing Verilog (log: designs/up_down_counter/runs/RUN_2026.06.07.08.58.39/logs/routing/19-global_write_netlist.log)...
[STEP 21]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/up_down_counter/runs/RUN_2026.06.07.08.58.39/logs/routing/21-grt_sta.log)...
[STEP 22]
[INFO]: Running Fill Insertion (log: designs/up_down_counter/runs/RUN_2026.06.07.08.58.39/logs/routing/22-fill.log)...
[STEP 23]
[INFO]: Running Detailed Routing (log: designs/up_down_counter/runs/RUN_2026.06.07.08.58.39/logs/routing/23-detailed.log)...
[INFO]: No DRC violations after detailed routing.
[STEP 24]
[INFO]: Checking Wire Lengths (log: designs/up_down_counter/runs/RUN_2026.06.07.08.58.39/logs/routing/24-wire_lengths.log)...
[STEP 25]
[INFO]: Running SPEF Extraction at the min process corner (log: designs/up_down_counter/runs/RUN_2026.06.07.08.58.39/logs/signoff/25-parasitics_extraction.min.log)...
[STEP 26]
```

mountain@mountain:~/OpenLane-Test-2/OpenLane-Test-2\$ cd /Users/mounika/OpenLane/designs/spm/runs/openlane_test/results/signoff/spm.magic.gds
/Users/mounika/OpenLane/designs/spm/runs/openlane_test/results/signoff/spm.gds
/Users/mounika/OpenLane/designs/spm/runs/openlane_test/results/signoff/spm.klayout.gds
/Users/mounika/OpenLane/designs/spm/runs/openlane_test/results/final/gds/spm.gds
/Users/mounika/OpenLane/designs/spm/runs/RUN_2026.06.07.08.20.54/results/signoff/spm.magic.gds
/Users/mounika/OpenLane/designs/spm/runs/RUN_2026.06.07.08.20.54/results/signoff/spm.gds
/Users/mounika/OpenLane/designs/spm/runs/RUN_2026.06.07.08.20.54/results/signoff/spm.klayout.gds
/Users/mounika/OpenLane/designs/spm/runs/RUN_2026.06.07.08.20.54/results/final/gds/spm.gds
/Users/mounika/OpenLane/designs/up_down_counter_32/runs/PART3_FINAL/results/signoff/up_down_counter_32.gds
/Users/mounika/OpenLane/designs/up_down_counter_32/runs/PART3_FINAL/results/signoff/up_down_counter_32.klayout.gds
/Users/mounika/OpenLane/designs/up_down_counter_32/runs/PART3_FINAL/results/signoff/up_down_counter_32.magic.gds
/Users/mounika/OpenLane/designs/up_down_counter_32/runs/PART3_FINAL/results/final/gds/up_down_counter_32.gds
/Users/mounika/OpenLane/designs/up_down_counter/runs/RUN_2026.06.06.13.09.54/results/signoff/up_down_counter.magic.gds
/Users/mounika/OpenLane/designs/up_down_counter/runs/RUN_2026.06.06.13.09.54/results/signoff/up_down_counter.klayout.gds
/Users/mounika/OpenLane/designs/up_down_counter/runs/RUN_2026.06.06.13.09.54/results/signoff/up_down_counter.gds
/Users/mounika/OpenLane/designs/up_down_counter/runs/RUN_2026.06.06.13.09.54/results/final/gds/up_down_counter.gds
/Users/mounika/OpenLane/designs/up_down_counter/runs/RUN_2026.06.06.05.13.44/results/signoff/up_down_counter.magic.gds
/Users/mounika/OpenLane/designs/up_down_counter/runs/RUN_2026.06.06.05.13.44/results/signoff/up_down_counter.klayout.gds
/Users/mounika/OpenLane/designs/up_down_counter/runs/RUN_2026.06.06.05.13.44/results/signoff/up_down_counter.gds
/Users/mounika/OpenLane/designs/up_down_counter/runs/RUN_2026.06.06.05.13.44/results/final/gds/up_down_counter.gds
/Users/mounika/OpenLane/designs/up_down_counter/runs/RUN_2026.06.06.05.07.43/results/signoff/up_down_counter.magic.gds
/Users/mounika/OpenLane/designs/up_down_counter/runs/RUN_2026.06.06.05.07.43/results/signoff/up_down_counter.klayout.gds
/Users/mounika/OpenLane/designs/up_down_counter/runs/RUN_2026.06.06.05.07.43/results/signoff/up_down_counter.gds
/Users/mounika/OpenLane/designs/up_down_counter/runs/RUN_2026.06.06.05.07.43/results/final/gds/up_down_counter.gds
/Users/mounika/OpenLane/designs/up_down_counter/runs/RUN_2026.06.06.04.57.42/results/signoff/up_down_counter.magic.gds
/Users/mounika/OpenLane/designs/up_down_counter/runs/RUN_2026.06.06.04.57.42/results/signoff/up_down_counter.klayout.gds
/Users/mounika/OpenLane/designs/up_down_counter/runs/RUN_2026.06.06.04.57.42/results/signoff/up_down_counter.gds
/Users/mounika/OpenLane/designs/up_down_counter/runs/RUN_2026.06.06.04.57.42/results/final/gds/up_down_counter.gds
/Users/mounika/OpenLane/designs/up_down_counter/runs/RUN_2026.06.06.14.17.33/results/signoff/up_down_counter.magic.gds
/Users/mounika/OpenLane/designs/up_down_counter/runs/RUN_2026.06.06.14.17.33/results/signoff/up_down_counter.klayout.gds
/Users/mounika/OpenLane/designs/up_down_counter/runs/RUN_2026.06.06.14.17.33/results/signoff/up_down_counter.gds
/Users/mounika/OpenLane/designs/up_down_counter/runs/RUN_2026.06.06.13.15.53/results/signoff/up_down_counter.magic.gds
/Users/mounika/OpenLane/designs/up_down_counter/runs/RUN_2026.06.06.13.15.53/results/signoff/up_down_counter.klayout.gds
/Users/mounika/OpenLane/designs/up_down_counter/runs/RUN_2026.06.06.13.15.53/results/signoff/up_down_counter.gds
/Users/mounika/OpenLane/designs/up_down_counter/runs/RUN_2026.06.06.13.15.53/results/final/gds/up_down_counter.gds
/Users/mounika/OpenLane/designs/up_down_counter/runs/RUN_2026.06.01.09.24.36/results/signoff/up_down_counter.magic.gds
/Users/mounika/OpenLane/designs/up_down_counter/runs/RUN_2026.06.01.09.24.36/results/signoff/up_down_counter.klayout.gds
/Users/mounika/OpenLane/designs/up_down_counter/runs/RUN_2026.06.01.09.24.36/results/signoff/up_down_counter.gds
/Users/mounika/OpenLane/designs/up_down_counter/runs/RUN_2026.06.01.09.24.36/results/final/gds/up_down_counter.gds
/Users/mounika/OpenLane/designs/up_down_counter/runs/RUN_2026.06.06.13.50.31/results/signoff/up_down_counter.magic.gds
/Users/mounika/OpenLane/designs/up_down_counter/runs/RUN_2026.06.06.13.50.31/results/signoff/up_down_counter.klayout.gds
/Users/mounika/OpenLane/designs/up_down_counter/runs/RUN_2026.06.06.13.50.31/results/signoff/up_down_counter.gds
/Users/mounika/OpenLane/designs/up_down_counter/runs/RUN_2026.06.06.13.50.31/results/final/gds/up_down_counter.gds
/Users/mounika/OpenLane/designs/up_down_counter/runs/RUN_2026.06.06.05.09.22/results/signoff/up_down_counter.magic.gds
/Users/mounika/OpenLane/designs/up_down_counter/runs/RUN_2026.06.06.05.09.22/results/signoff/up_down_counter.klayout.gds
/Users/mounika/OpenLane/designs/up_down_counter/runs/RUN_2026.06.06.05.09.22/results/signoff/up_down_counter.gds
/Users/mounika/OpenLane/designs/up_down_counter/runs/RUN_2026.06.06.05.09.22/results/final/gds/up_down_counter.gds
/Users/mounika/OpenLane/designs/up_down_counter/runs/RUN_2026.06.06.14.14.38/results/signoff/up_down_counter.magic.gds
/Users/mounika/OpenLane/designs/up_down_counter/runs/RUN_2026.06.06.14.14.38/results/signoff/up_down_counter.klayout.gds
/Users/mounika/OpenLane/designs/up_down_counter/runs/RUN_2026.06.06.14.14.38/results/signoff/up_down_counter.gds
/Users/mounika/OpenLane/designs/up_down_counter/runs/RUN_2026.06.06.14.14.38/results/final/gds/up_down_counter.gds
/Users/mounika/OpenLane/designs/up_down_counter/runs/RUN_2026.06.06.05.21.42/results/signoff/up_down_counter.magic.gds
/Users/mounika/OpenLane/designs/up_down_counter/runs/RUN_2026.06.06.05.21.42/results/signoff/up_down_counter.klayout.gds
/Users/mounika/OpenLane/designs/up_down_counter/runs/RUN_2026.06.06.05.21.42/results/signoff/up_down_counter.gds
/Users/mounika/OpenLane/designs/up_down_counter/runs/RUN_2026.06.06.05.21.42/results/final/gds/up_down_counter.gds
/Users/mounika/OpenLane/up_down_counter.gds



Part 3: 32-bit Up/Down Counter Custom Run:

Parameter	Required Value
Counter	32-bit
Clock Period	6 ns
Setup Uncertainty	10%
Derate	3%
Utilization	70%
Tap Columns	2
Core Ring	Enable
CTS Skew	5%

```
openlane -config /ff509f65b17bfa4068d5336495ab1718987ff69 -flow t1 -design up_down_counter_32 -tag PART3_FINAL -overwrite
openlane v1.0.2 (ff509f65b17bfa4068d5336495ab1718987ff69)
All rights reserved. (c) 2020-2025 Efabless Corporation and contributors.
Available under the Apache License, version 2.0. See the LICENSE file for more details.

INFO: Using configuration in 'designs/up_down_counter_32/config.json'...
INFO: PDK Root: /Users/mounika/.ciel
INFO: Process Design Kit: sky130A
INFO: Standard Cell Library: sky130_fd_sc_hd
INFO: Optimization Standard Cell Library: sky130_fd_sc_hd
INFO: Run Directory: /openlane/designs/up_down_counter_32/runs/PART3_FINAL
INFO: Removing existing /openlane/designs/up_down_counter_32/runs/PART3_FINAL...
INFO: Saving runtime environment...
INFO: Preparing LEF files for the nom corner...
INFO: Preparing LEF files for the min corner...
INFO: Preparing LEF files for the max corner...
INFO: Running linter (Verilator) (log: designs/up_down_counter_32/runs/PART3_FINAL/logs/synthesis/linter.log)...
INFO: 0 errors found by linter
INFO: 0 warnings found by linter
STEP 11
INFO: Running Synthesis (log: designs/up_down_counter_32/runs/PART3_FINAL/logs/synthesis/1-synthesis.log)...
STEP 21
INFO: Running Single-Corner Static Timing Analysis (log: designs/up_down_counter_32/runs/PART3_FINAL/logs/synthesis/2-sta.log)...
STEP 31
INFO: Running Initial Floorplanning (log: designs/up_down_counter_32/runs/PART3_FINAL/logs/floorplan/3-initial_fp.log)...
WARNING: Current core area is too small for the power grid settings chosen. The power grid will be scaled down.
INFO: Floorplanned with width 61.64 and height 59.84.
STEP 41
INFO: Running IO Placement...
STEP 51
INFO: Running Tap/Decap Insertion (log: designs/up_down_counter_32/runs/PART3_FINAL/logs/floorplan/5-tap.log)...
INFO: Power planning with power {VPWR} and ground {VGND}...
STEP 61
INFO: Generating PDN (log: designs/up_down_counter_32/runs/PART3_FINAL/logs/floorplan/6-pdn.log)...
STEP 71
INFO: Running Global Placement (log: designs/up_down_counter_32/runs/PART3_FINAL/logs/placement/7-global.log)...
STEP 81
INFO: Running Single-Corner Static Timing Analysis (log: designs/up_down_counter_32/runs/PART3_FINAL/logs/placement/8-gpl_sta.log)...
STEP 91
INFO: Running Placement Resizer Design Optimizations (log: designs/up_down_counter_32/runs/PART3_FINAL/logs/placement/9-resizer.log)...
STEP 101
INFO: Running Detailed Placement (log: designs/up_down_counter_32/runs/PART3_FINAL/logs/placement/10-detailed.log)...
STEP 111
INFO: Running Single-Corner Static Timing Analysis (log: designs/up_down_counter_32/runs/PART3_FINAL/logs/placement/11-dpl_sta.log)...
STEP 121
INFO: Running Clock Tree Synthesis (log: designs/up_down_counter_32/runs/PART3_FINAL/logs/cts/12-cts.log)...
STEP 131
INFO: Running Single-Corner Static Timing Analysis (log: designs/up_down_counter_32/runs/PART3_FINAL/logs/cts/13-cts_sta.log)...
STEP 141
INFO: Running Placement Resizer Timing Optimizations (log: designs/up_down_counter_32/runs/PART3_FINAL/logs/cts/14-resizer.log)...
STEP 151
INFO: Running Global Routing Resizer Design Optimizations (log: designs/up_down_counter_32/runs/PART3_FINAL/logs/routing/15-resizer_design.log)...
STEP 161
```

```
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/up_down_counter_32/runs/PART3_FINAL/logs/routing/16-rsz_design_sta.log)...\n[STEP 17]\n[INFO]: Running Global Routing Resizer Timing Optimizations (log: designs/up_down_counter_32/runs/PART3_FINAL/logs/routing/17-resizer_timing.log)...\n[STEP 18]\n[INFO]: Running Single-Corner Static Timing Analysis (log: designs/up_down_counter_32/runs/PART3_FINAL/logs/routing/18-rsz_timing_sta.log)...\n[STEP 19]\n[INFO]: Running Global Routing (log: designs/up_down_counter_32/runs/PART3_FINAL/logs/routing/19-global.log)...\n[INFO]: Starting OpenROAD Antenna Repair Iterations...\n[STEP 20]\n[INFO]: Writing Verilog (log: designs/up_down_counter_32/runs/PART3_FINAL/logs/routing/19-global_write_netlist.log)...\n[STEP 21]\n[INFO]: Running Single-Corner Static Timing Analysis (log: designs/up_down_counter_32/runs/PART3_FINAL/logs/routing/21-grt_sta.log)...\n[STEP 22]\n[INFO]: Running Fill Insertion (log: designs/up_down_counter_32/runs/PART3_FINAL/logs/routing/22-fill.log)...\n[STEP 23]\n[INFO]: Running Detailed Routing (log: designs/up_down_counter_32/runs/PART3_FINAL/logs/routing/23-detailed.log)...\n[INFO]: No DRC violations after detailed routing.\n[STEP 24]\n[INFO]: Checking Wire Lengths (log: designs/up_down_counter_32/runs/PART3_FINAL/logs/routing/24-wire_lengths.log)...\n[STEP 25]\n[INFO]: Running SPEF Extraction at the min process corner (log: designs/up_down_counter_32/runs/PART3_FINAL/logs/signoff/25-parasitics_extraction.min.log)...\n[STEP 26]\n[INFO]: Running Multi-Corner Static Timing Analysis at the min process corner (log: designs/up_down_counter_32/runs/PART3_FINAL/logs/signoff/26-rcx_mcsta.min.log)...\n[STEP 27]\n[INFO]: Running SPEF Extraction at the max process corner (log: designs/up_down_counter_32/runs/PART3_FINAL/logs/signoff/27-parasitics_extraction.max.log)...\n[STEP 28]\n[INFO]: Running Multi-Corner Static Timing Analysis at the max process corner (log: designs/up_down_counter_32/runs/PART3_FINAL/logs/signoff/28-rcx_mcsta.max.log)...\n[STEP 29]\n[INFO]: Running SPEF Extraction at the nom process corner (log: designs/up_down_counter_32/runs/PART3_FINAL/logs/signoff/29-parasitics_extraction.nom.log)...\n[STEP 30]\n[INFO]: Running Multi-Corner Static Timing Analysis at the nom process corner (log: designs/up_down_counter_32/runs/PART3_FINAL/logs/signoff/30-rcx_mcsta.nom.log)...\n[STEP 31]\n[INFO]: Running Single-Corner Static Timing Analysis at the nom process corner (log: designs/up_down_counter_32/runs/PART3_FINAL/logs/signoff/31-rcx_sta.log)...\n[WARNING]: Module sky130_fd_sc_hd__tapwvrvrpd_1 blackboxed during sta\n[WARNING]: Module sky130_fd_sc_hd__fill_2 blackboxed during sta\n[WARNING]: Module sky130_fd_sc_hd__fill_1 blackboxed during sta\n[WARNING]: Module sky130_ef_sc_hd__decap_12 blackboxed during sta\n[STEP 32]\n[INFO]: Creating IR Drop Report (log: designs/up_down_counter_32/runs/PART3_FINAL/logs/signoff/32-irdrop.log)...\n[WARNING]: VSRC_LOC_FILES is not defined. The IR drop analysis will run, but the values may be inaccurate.\n[STEP 33]\n[INFO]: Running Magic to generate various views...\n[INFO]: Streaming out GDSII with Magic (log: designs/up_down_counter_32/runs/PART3_FINAL/logs/signoff/33-gdsii.log)...\n[INFO]: Generating MAGLEF views...\n[INFO]: Generating lef with Magic (/openlane/designs/up_down_counter_32/runs/PART3_FINAL/logs/signoff/33-lef.log)...\n[STEP 34]\n[INFO]: Streaming out GDSII with KLayout (log: designs/up_down_counter_32/runs/PART3_FINAL/logs/signoff/34-gdsii-klayout.log)...\n[STEP 35]\n[INFO]: Running XOR on the layouts using KLayout (log: designs/up_down_counter_32/runs/PART3_FINAL/logs/signoff/35-xor.log)...\n[INFO]: No XOR differences between KLayout and Magic gds.\n[STEP 36]\n[INFO]: Running Magic Spice Export from LEF (log: designs/up_down_counter_32/runs/PART3_FINAL/logs/signoff/36-spice.log)...\n[STEP 37]\n[INFO]: Writing Powered Verilog (logs: designs/up_down_counter_32/runs/PART3_FINAL/logs/signoff/37-write_powered_def.log, designs/up_down_counter_32/runs/PART3_FINAL/logs/signoff/37-write_powered_verilog.log)\n\n[STEP 38]\n[INFO]: Writing Verilog (log: designs/up_down_counter_32/runs/PART3_FINAL/logs/signoff/37-write_powered_verilog.log)...\n[STEP 39]\n[INFO]: Running LVS (log: designs/up_down_counter_32/runs/PART3_FINAL/logs/signoff/39-lvs.lef.log)...\n[STEP 40]\n[INFO]: Running Magic DRC (log: designs/up_down_counter_32/runs/PART3_FINAL/logs/signoff/40-drc.log)...\n[INFO]: Converting Magic DRC database to various tool-readable formats...
```

```
[INFO]: Running OpenROAD Antenna Rule Checker (log: designs/up_down_counter_32/runs/PART3_FINAL/logs/signoff/41-arc.log)...\n[STEP 42]\n[INFO]: Running Circuit Validity Checker ERC (log: designs/up_down_counter_32/runs/PART3_FINAL/logs/signoff/42-erc_screen.log)...\n[INFO]: Saving current set of views in 'designs/up_down_counter_32/runs/PART3_FINAL/results/final'...\n[INFO]: Saving runtime environment...\n[INFO]: Generating final set of reports...\n[INFO]: Created manufacturability report at 'designs/up_down_counter_32/runs/PART3_FINAL/reports/manufacturability.rpt'.  
[INFO]: Created metrics report at 'designs/up_down_counter_32/runs/PART3_FINAL/reports/metrics.csv'.  
[INFO]: There are no max slew, max fanout or max capacitance violations in the design at the typical corner.  
[INFO]: There are no hold violations in the design at the typical corner.  
[INFO]: There are no setup violations in the design at the typical corner.  
[SUCCESS]: Flow complete.
```

```

up_down_counter_32.def up_down_counter_32.odt up_down_counter_32.sdc

./runs/PART3_RUN/results/floorplan:
up_down_counter_32.def up_down_counter_32.odt

./runs/PART3_RUN/results/placement:
up_down_counter_32.def up_down_counter_32.odt
up_down_counter_32.nl.v up_down_counter_32.pnl.v

./runs/PART3_RUN/results/routing:
up_down_counter_32.def up_down_counter_32.odt
up_down_counter_32.nl.v up_down_counter_32.pnl.v

./runs/PART3_RUN/results/signoff:
up_down_counter_32.lib up_down_counter_32.sdf

./runs/PART3_RUN/results/synthesis:
up_down_counter_32.sdf up_down_counter_32.v

./runs/PART3_RUN/tmp:
14-up_down_counter_32.def 16-up_down_counter_32.odt merged.max.lef
14-up_down_counter_32.nl.v 16-up_down_counter_32.pnl.v merged.min.lef
14-up_down_counter_32.odt 16-up_down_counter_32.sdc merged.nom.lef
14-up_down_counter_32.pnl.v cts placement
14-up_down_counter_32.sdc dimensions.txt routing
16-up_down_counter_32.def floorplan signoff
16-up_down_counter_32.nl.v layers.list synthesis

./runs/PART3_RUN/tmp/cts:
cts-fastest.lib cts.lib.exclude.list
cts-fastest.lib.exclude.list cts-slowest.lib
cts.lib cts-slowest.lib.exclude.list

./runs/PART3_RUN/tmp/floorplan:
3-initial_fp.def 3-initial_fp.sdc 4-io.def 5-tapcell.def
3-initial_fp.odt 3-minimized_pdn.txt 4-io.odt 5-tapcell.odt

./runs/PART3_RUN/tmp/placement:
7-global.def 7-global.odt 9-resizer.def 9-resizer.odt 9-resizer.sdc
7-global.nl.v 7-global.pnl.v 9-resizer.nl.v 9-resizer.pnl.v

./runs/PART3_RUN/tmp/routing:
18-global.def 21-fill.odt groute-congestion.rpt
18-global.guide 21-fill.pnl.v resizer-routing-design-congestion.rpt
18-global.odt config.tracks resizer-routing-timing-congestion.rpt
21-fill.def global.nl.v
21-fill.nl.v global.pnl.v

./runs/PART3_RUN/tmp/signoff:

./runs/PART3_RUN/tmp/synthesis:
1-sky130_fd_sc_hd__tt_025C_1v80.no_pg.lib post_techmap.dot
1-trimmed.no_pg.lib synthesis.sdc
hierarchy.dot trimmed.lib
merged.lib trimmed.lib.exclude.list

./runs/RUN_2026.06.04_13.54.12:
config_in.tcl

./runs/RUN_2026.06.04_14.06.24:
config_in.tcl

./src:
up_down_counter_32.v

```

```

OpenLane Container (ff5509f):/openlane/designs/up_down_counter_32$

```

```

openlane Container (ff5699f):openlane$ grep -s clock /openlane/designs/up_down_counter_32/config.json
"CLOCK_PORT": "clk",
"CLOCK_PERIOD": 6,
"SYNTH_CLOCK_UNCERTAINTY": 0.6,
openlane Container (ff5699f):openlane$ grep -R "UNCERTAINTY" /openlane/designs/up_down_counter_32
/openlane/designs/up_down_counter_32/config.json:  "SYNTH_CLOCK_UNCERTAINTY": 0.6,
/openlane/designs/up_down_counter_32/runs/PART3_FINAL/config.tcl:set ::env(SYNTH_CLOCK_UNCERTAINTY) "0.6"
openlane Container (ff5699f):openlane$ grep -R "DERATE" /openlane/designs/up_down_counter_32
/openlane/designs/up_down_counter_32/config.json:  "SYNTH_TIMING_DERATE": 0.03,
/openlane/designs/up_down_counter_32/runs/PART3_FINAL/config.tcl:set ::env(SYNTH_TIMING_DERATE) "0.03"
openlane/designs/up_down_counter_32/runs/PART3_RUN/config.tcl:set ::env(SYNTH_TIMING_DERATE) "0.03"
openlane Container (ff5699f):openlane$ grep -R "FP_CORE_UTIL" /openlane/designs/up_down_counter_32
/openlane/designs/up_down_counter_32/config.json:  "FP_CORE_UTIL": 70,
/openlane/designs/up_down_counter_32/runs/PART3_FINAL/config.tcl:set ::env(FP_CORE_UTIL) "70"
/openlane/designs/up_down_counter_32/runs/PART3_FINAL/reports/metrics.csv:design,design_name,config,flow_status,total_runtime,routed_runtime,(Cell/mm^2)/Core_Util,DIEAREA_mm^2,CellPer_mm^2,OpenDP_Util,Final_U
til,Peak_Memory_Usage_MB,synth_cell_count,tritonRoute_violations,Short_violations,NetSpc_violations,Offgrid_violations,MinHole_violations,Other_violations,Magic_violations,pin_antenna_violations,net_antenna_v
iolations,lvs_total_errors,rcv_total_errors,layout_violations,wire_length,vias,wms,pl_wms,optimized_wms,fastroute_wms,spef_wms,tns,pl_tns,optimized_tns,fastroute_tns,spef_tns,HPWL,routing_layer1_pct,routing_
layer_pct,routing_layer3_pct,routing_layer4_pct,routing_layer5_pct,routing_layer6_pct,wires_count,wire_bits,public_wires_count,public_wire_bits,memory_bits,processes_count,cells_pre_abc,AND_DF
F,MAND,NOR_OR,XOR,XNOR,MUX,inputs,outputs_level,DecapCells,WelltapCells,DiodeCells,FillCells,NonPhysCells,TotalCells,CoreArea_mm^2,power_slowest_internal_uW,power_slowest_switching_uW,power_slowest_leakage_uW
,power_typical_internal_uW,power_typical_switching_uW,power_typical_leakage_uW,power_fastest_internal_uW,power_fastest_switching_uW,power_fastest_leakage_uW,critical_path_ns,suggested_clock_period,suggested_c
lock_frequency,CLOCK_PERIOD,FP_ASPECT_RATIO,FP_CORE_UTIL,FP_PDN_HPITCH,FP_PDN_VPITCH,GR1_ADJUSTMENT,GR1_REPAIR_ANTENNAS,MAX_FANOUT_CONSTRAINT,PL_TARGET_DENSITY,RUN_HEURISTIC_DIODE_INSERTION,STD_CELL_LIBRARY,S
YNTH_STRATEGY
/openlane/designs/up_down_counter_32/runs/PART3_RUN/config.tcl:set ::env(FP_CORE_UTIL) "70"
openlane Container (ff5699f):openlane$ grep -i tap /openlane/designs/up_down_counter_32/runs/PART3_RUN/logs/floorplan/*.log
/openlane/designs/up_down_counter_32/runs/PART3_RUN/logs/floorplan/5-tap.log:[INFO TAP-0004] Inserted 44 endcaps.
/openlane/designs/up_down_counter_32/runs/PART3_RUN/logs/floorplan/5-tap.log:[INFO TAP-0005] Inserted 48 tapcells.
/openlane/designs/up_down_counter_32/runs/PART3_RUN/logs/floorplan/5-tap.log:Writing OpenROAD database to '/openlane/designs/up_down_counter_32/runs/PART3_RUN/tmp/floorplan/5-tapcell.odb'-
/openlane/designs/up_down_counter_32/runs/PART3_RUN/logs/floorplan/6-pdn.log:[INFO]: Reading ODB at '/openlane/designs/up_down_counter_32/runs/PART3_RUN/tmp/floorplan/5-tapcell.odb'-
openlane Container (ff5699f):openlane$ grep -R "CORE_RING" /openlane/designs/up_down_counter_32
/openlane/designs/up_down_counter_32/config.json:  "FP_PDN_CORE_RING": true,
/openlane/designs/up_down_counter_32/runs/PART3_FINAL/config.tcl:set ::env(FP_PDN_CORE_RING) "1"
/openlane/designs/up_down_counter_32/runs/PART3_FINAL/config.tcl:set ::env(FP_PDN_CORE_RING_HOFFSET) "6"
/openlane/designs/up_down_counter_32/runs/PART3_FINAL/config.tcl:set ::env(FP_PDN_CORE_RING_HSPACING) "1.7"
/openlane/designs/up_down_counter_32/runs/PART3_FINAL/config.tcl:set ::env(FP_PDN_CORE_RING_HWIDTH) "1.6"
/openlane/designs/up_down_counter_32/runs/PART3_FINAL/config.tcl:set ::env(FP_PDN_CORE_RING_VOFFSET) "6"
/openlane/designs/up_down_counter_32/runs/PART3_FINAL/config.tcl:set ::env(FP_PDN_CORE_RING_VSPACING) "1.7"
/openlane/designs/up_down_counter_32/runs/PART3_FINAL/config.tcl:set ::env(FP_PDN_CORE_RING_VWIDTH) "1.6"
/openlane/designs/up_down_counter_32/runs/PART3_RUN/config.tcl:set ::env(FP_PDN_CORE_RING) "1"
/openlane/designs/up_down_counter_32/runs/PART3_RUN/config.tcl:set ::env(FP_PDN_CORE_RING_HOFFSET) "6"
/openlane/designs/up_down_counter_32/runs/PART3_RUN/config.tcl:set ::env(FP_PDN_CORE_RING_HSPACING) "1.7"
/openlane/designs/up_down_counter_32/runs/PART3_RUN/config.tcl:set ::env(FP_PDN_CORE_RING_HWIDTH) "1.6"
/openlane/designs/up_down_counter_32/runs/PART3_RUN/config.tcl:set ::env(FP_PDN_CORE_RING_VOFFSET) "6"
/openlane/designs/up_down_counter_32/runs/PART3_RUN/config.tcl:set ::env(FP_PDN_CORE_RING_VSPACING) "1.7"
/openlane/designs/up_down_counter_32/runs/PART3_RUN/config.tcl:set ::env(FP_PDN_CORE_RING_VWIDTH) "1.6"
openlane Container (ff5699f):openlane$ grep -R "SKEW" /openlane/designs/up_down_counter_32
/openlane/designs/up_down_counter_32/config.json:  "PL_TARGET_SKEW": 0.3
/openlane/designs/up_down_counter_32/runs/PART3_FINAL/config.tcl:set ::env(PL_TARGET_SKEW) "0.3"

```

```

openlane Container (ff5699f):openlane$ grep "FP_PDN_CORE_RING" /openlane/designs/up_down_counter_32/runs/PART3_FINAL/config.tcl
set ::env(FP_PDN_CORE_RING) "1"
set ::env(FP_PDN_CORE_RING_HOFFSET) "6"
set ::env(FP_PDN_CORE_RING_HSPACING) "1.7"
set ::env(FP_PDN_CORE_RING_HWIDTH) "1.6"
set ::env(FP_PDN_CORE_RING_VOFFSET) "6"
set ::env(FP_PDN_CORE_RING_VSPACING) "1.7"
set ::env(FP_PDN_CORE_RING_VWIDTH) "1.6"
openlane Container (ff5699f):openlane$

```